2.3 $\frac{x^2}{x-1} - \frac{1}{x-1} = \frac{x^2-1}{x-1}$

19 $\lim_{x\to 1} \underbrace{\left(\frac{x^2}{x-1} - \frac{1}{x-1}\right)}_{f(x)} =$

$x \to 1 \Rightarrow x \approx 1$

$f(x) \longrightarrow \frac{1^2}{1-1} - \frac{1}{1-1} = \frac{1}{0} - \frac{1}{0}$

$\underbrace{\infty - \infty}$

!? INDET.

$f(x) = \frac{x^2 - 1^2}{x-1} = \frac{(x+1)\cancel{(x-1)}}{\cancel{(x-1)}} = (x+1)$

$a^2 - b^2 = (a+b)(a-b)$

$\lim_{x\to 1} f(x) = \lim_{x\to 1} (x+1) = 2$ //

26 $\lim_{h\to 0} \overbrace{\left(\frac{1}{h}\right)\left(\frac{1}{\sqrt{1+h}} - 1\right)}^{f(h)}$

$h \to 0 \Rightarrow h \approx 0 \Rightarrow f(h) \longrightarrow \frac{1}{0} \cdot \Big(\overbrace{\frac{1}{\sqrt{1+0}} - 1}^{0}\Big)$

$0/0$ (!?) INDET.

TRATAMENTO ALGÉBRICO:

$\frac{1}{h} \cdot \left(\frac{1}{\sqrt{1+h}} - \frac{1 \times \sqrt{1+h}}{1 \times \sqrt{1+h}}\right) = \frac{1}{h} \cdot \left(\frac{1 - \sqrt{1+h}}{\sqrt{1+h}}\right) \cdot \left(\frac{1+\sqrt{1+h}}{1+\sqrt{1+h}}\right)$

$$= \frac{1}{h} \cdot \left(\frac{\cancel{1} + \cancel{\sqrt{1+h}} - \cancel{\sqrt{1+h}} - (\cancel{1}+h)}{\sqrt{1+h}\cdot\left(1+\sqrt{1+h}\right)} \right)$$

$$= \frac{1}{\cancel{h}} \cdot \left(\frac{-\cancel{h}}{\sqrt{1+h}\cdot\left(1+\sqrt{1+h}\right)} \right) = \overbrace{\frac{-1}{\underbrace{\sqrt{1+h}}_{1}\cdot\left(\underbrace{1+\underbrace{\sqrt{1+h}}_{1}}_{2}\right)}}^{f(x)}$$

$h \to 0 \Rightarrow h \approx 0$

$$\lim_{h\to 0} \left(\frac{-1}{\sqrt{1+h}\cdot\left(1+\sqrt{1+h}\right)} \right) = \frac{-1}{1\cdot 2} = -1/2$$

27 $\displaystyle \lim_{x\to 1} \overbrace{\frac{(x-1)^5}{x^5-1}}^{f(x)}$

$$f(x) = \frac{(x-1)\cdot(x-1)\cdot(x-1)\cdot(x-1)\cdot(x-1)}{x^5-1}$$

1 é raiz $\Rightarrow$ então $(x-1)$ divide (x^5-1)

$\underbrace{x^5+0x^4+0x^3+0x^2}+0x-1$

$$\begin{array}{l|l}
x^5 - 1 & x-1 \\ \hline
-x^5 + x^4 & x^4+x^3+x^2+x+1 \\ \hline
x^4 - 1 & \\
-x^4 + x^3 & \\ \hline
x^3 - 1 & \\
-x^3 + x^2 & \\ \hline
x^2 - 1 & \\
-x^2 + x & \\ \hline
x - 1 & \\
-x + 1 & \\ \hline
0 &
\end{array}$$

Logo, $x^5-1 = (x-1)\cdot(x^4+x^3+x^2+x+1)$

Portanto,

$$f(x) = \frac{\cancel{(x-1)}\cdot(x-1)\cdot(x-1)\cdot(x-1)\cdot(x-1)}{\cancel{(x-1)}(x^4+x^3+x^2+x+1)} = \frac{(x-1)^4}{x^4+x^3+x^2+x+1}$$

ENTÃO,

$$\lim_{x \to 1} f(x) = \lim_{x \to 1} \left(\frac{(x-1)^4}{x^4+x^3+x^2+x+1} \right) = \frac{0}{5} = 0$$

$$33 \quad \underbrace{\lim_{x \to 9} \frac{x^2 - 81}{3 - \sqrt{x}}}_{f(x)} = \frac{(x+9)(x-9)}{3-\sqrt{x}}$$

$$x - 9 = (\sqrt{x})^2 - (3)^2 = (\sqrt{x}+3)(\sqrt{x}-3)$$

$$= \frac{(x+9).(\sqrt{x}+3).\cancel{(\sqrt{x}-3)}}{-\cancel{(\sqrt{x}-3)}} = \frac{(x+9)(\sqrt{x}+3)}{-1}$$

$$\lim_{x \to 9} f(x) = \frac{(9+9).(\sqrt{9}+3)}{-1} = \frac{18.6}{-1} = -108$$

$$36 \quad \lim_{h \to 0} \frac{(9+h)^{-1} - 9^{-1}}{h} \quad f(x)$$

$$\frac{\frac{1}{9+h} - \frac{1}{9}}{h} = \frac{\frac{\cancel{9} - (\cancel{9}+h)}{9(9+h)}}{\frac{h}{1}} = \frac{-\cancel{h}}{9.(9+h).\cancel{h}}$$

$$= -\frac{1}{9.(9+h)} \quad \lim_{h \to 0} f(x) = -\frac{1}{81}$$